Shaheed Zulfikar Ali Bhutto Institute of Science & Technology University

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

Total Marks: 2.5

Obtained Marks: _____

Artificial Neural Network (Lab)

Task no # 01

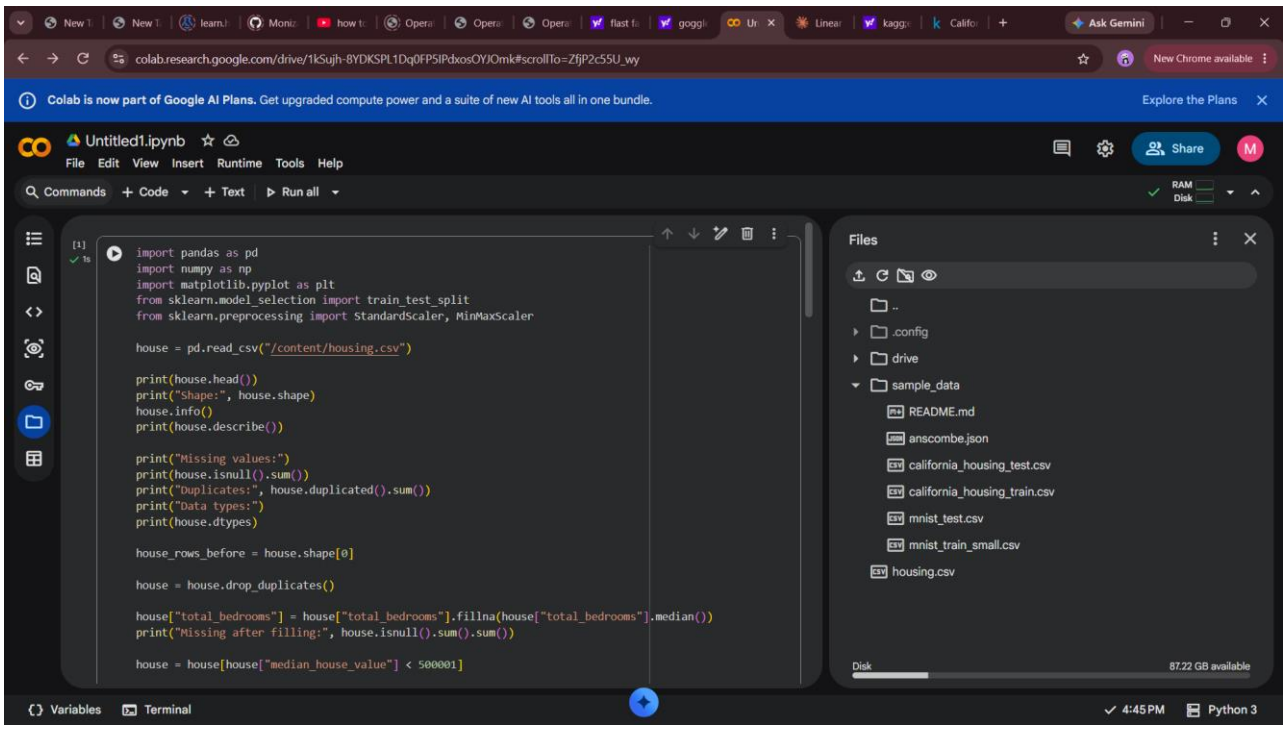
Last date of Submission: 13th October, 2026

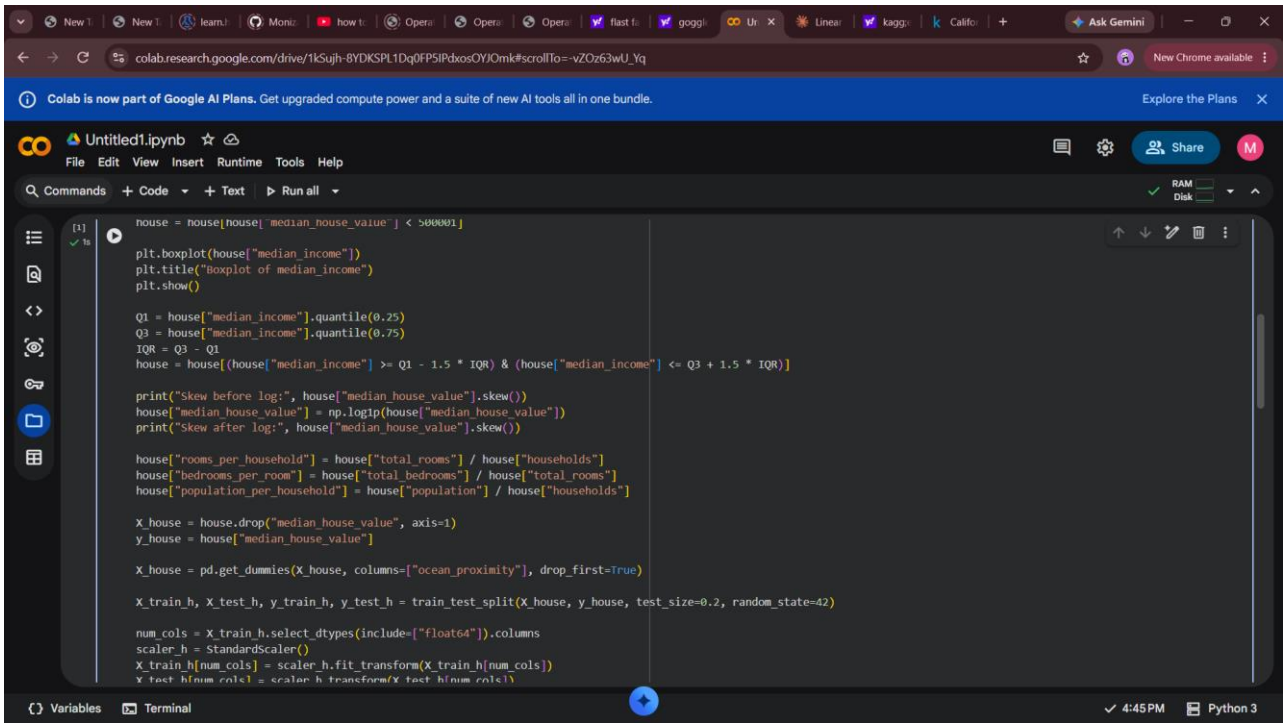
Submitted To: Muhammad Hassan Mujtaba

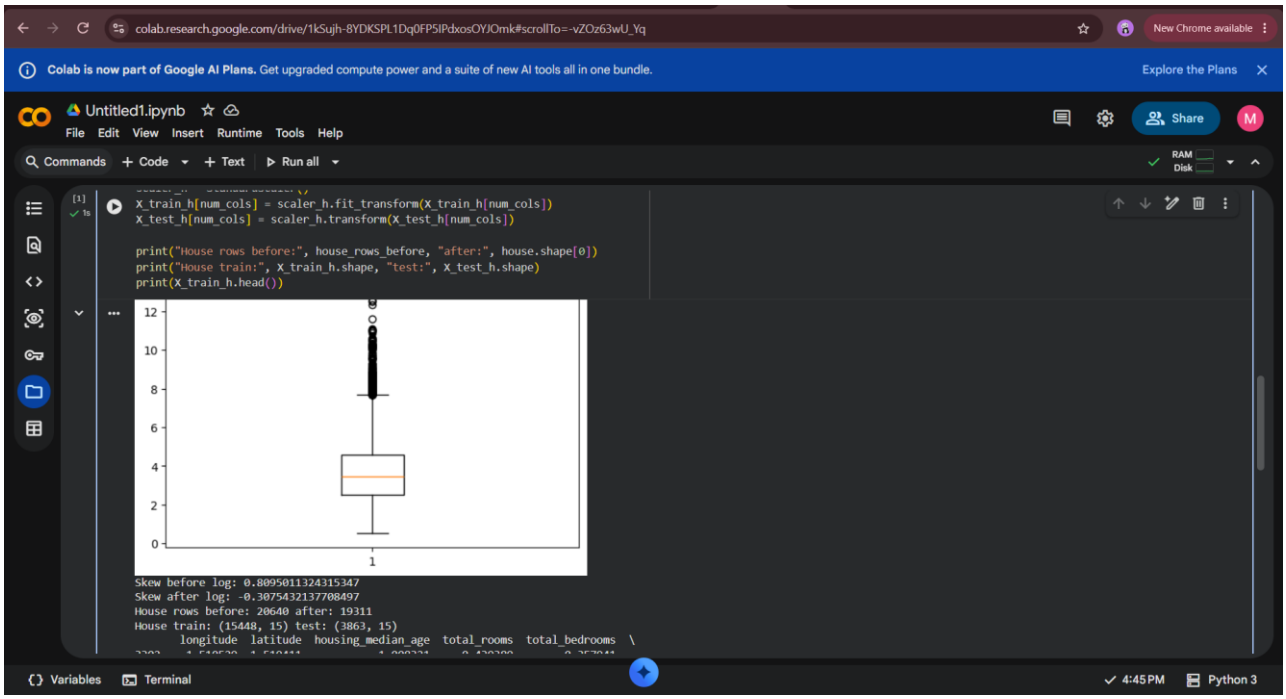
Student Name: Urooj Fatima

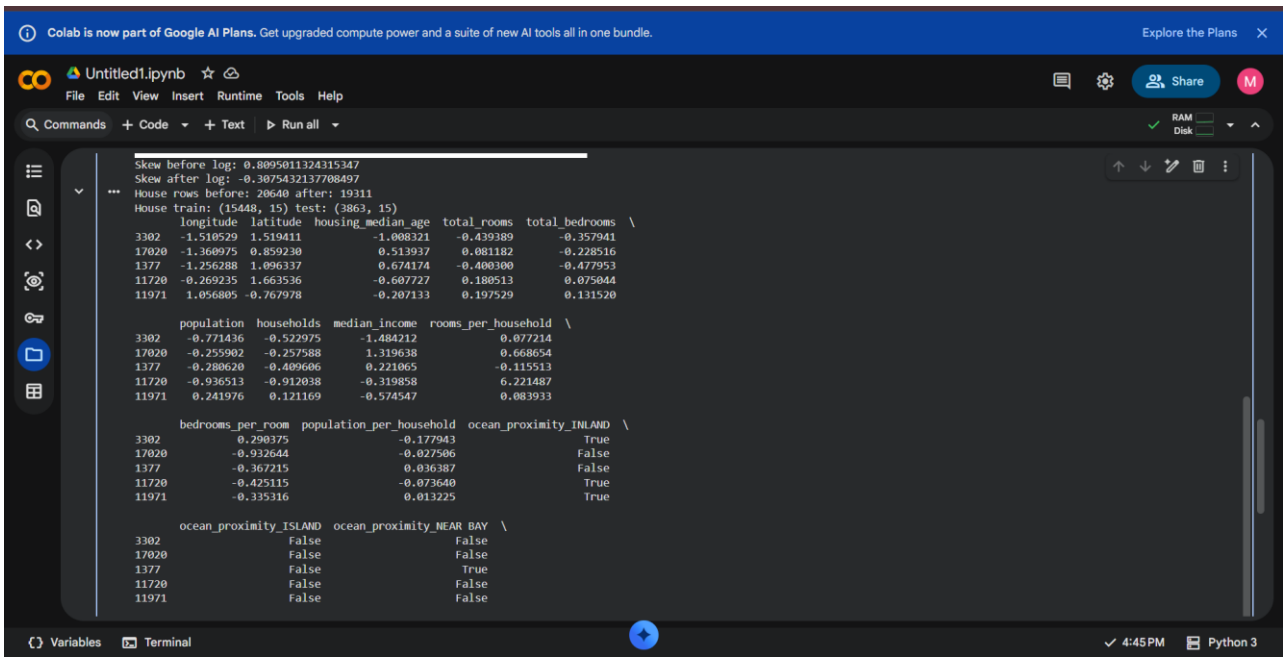
Registration Number: 24108238

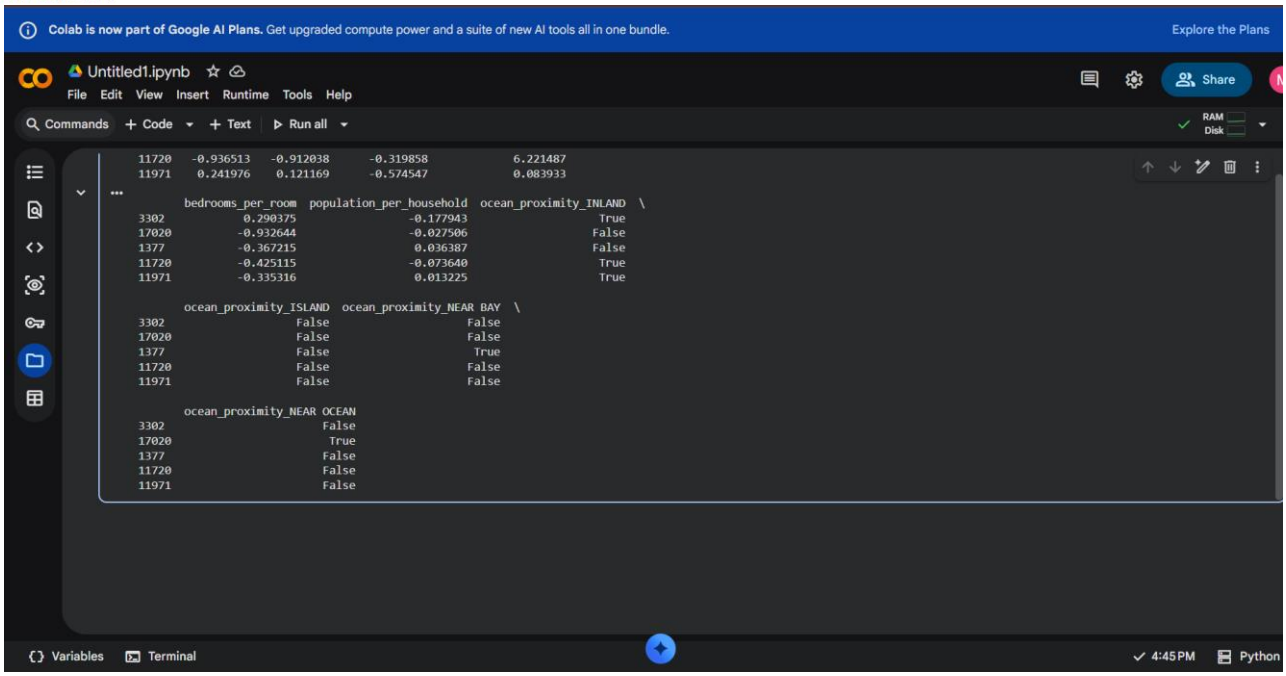
House Price Prediction System

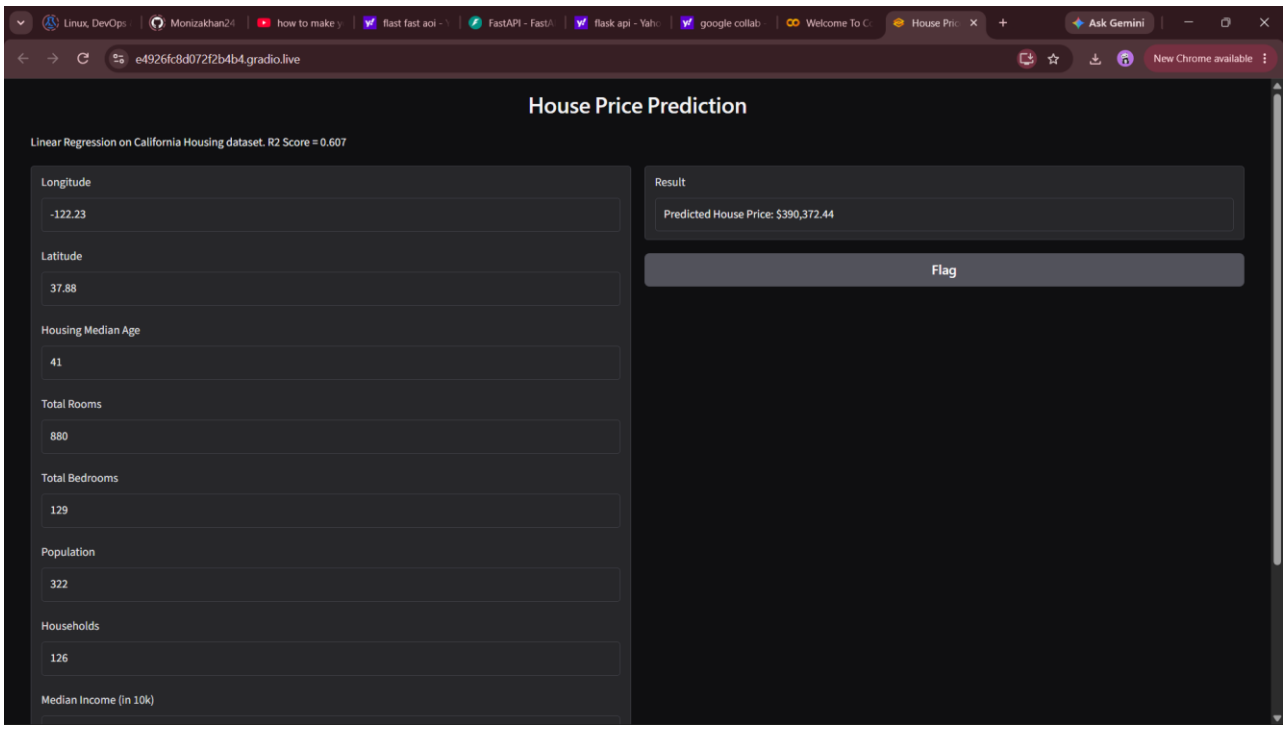


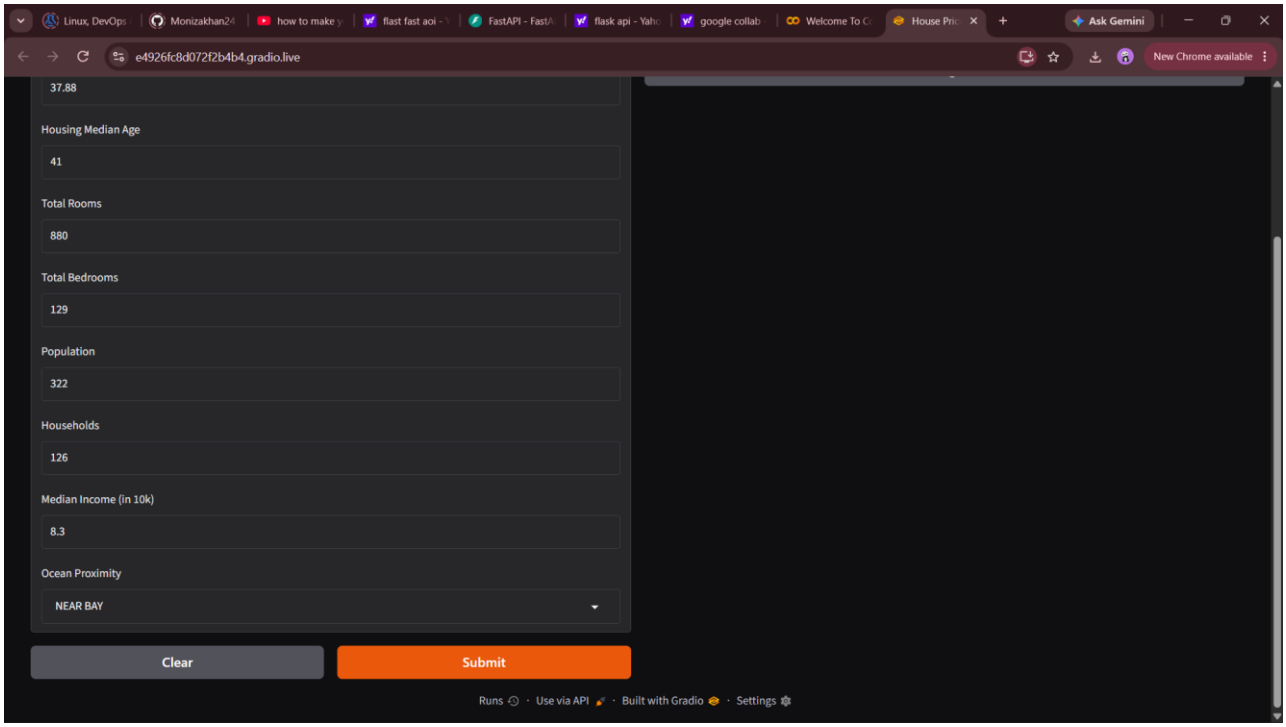



colab.research.google.com/drive/1kSujh-8YDKSPL1Dq0FP5IPdxosOYJOmk#scrollTo=-vZOz63wU_Yq
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RAM Disk
X_train_h[num_cols] = scaler_h.fit_transform(X_train_h[num_cols])
X_test_h[num_cols] = scaler_h.transform(X_test_h[num_cols])
print("House rows before:", house_rows_before, "after:", house.shape[0])
print("House train:", X_train_h.shape, "test:", X_test_h.shape)
print(X_train_h.head())
Skew before log: 0.8095011324315347
Skew after log: -0.3075432137708497
House rows before: 20640 after: 19311
House train: (15448, 15) test: (3863, 15)
longitude latitude housing_median_age total_rooms total_bedrooms \
Variables
Terminal
4:45 PM
Python 3

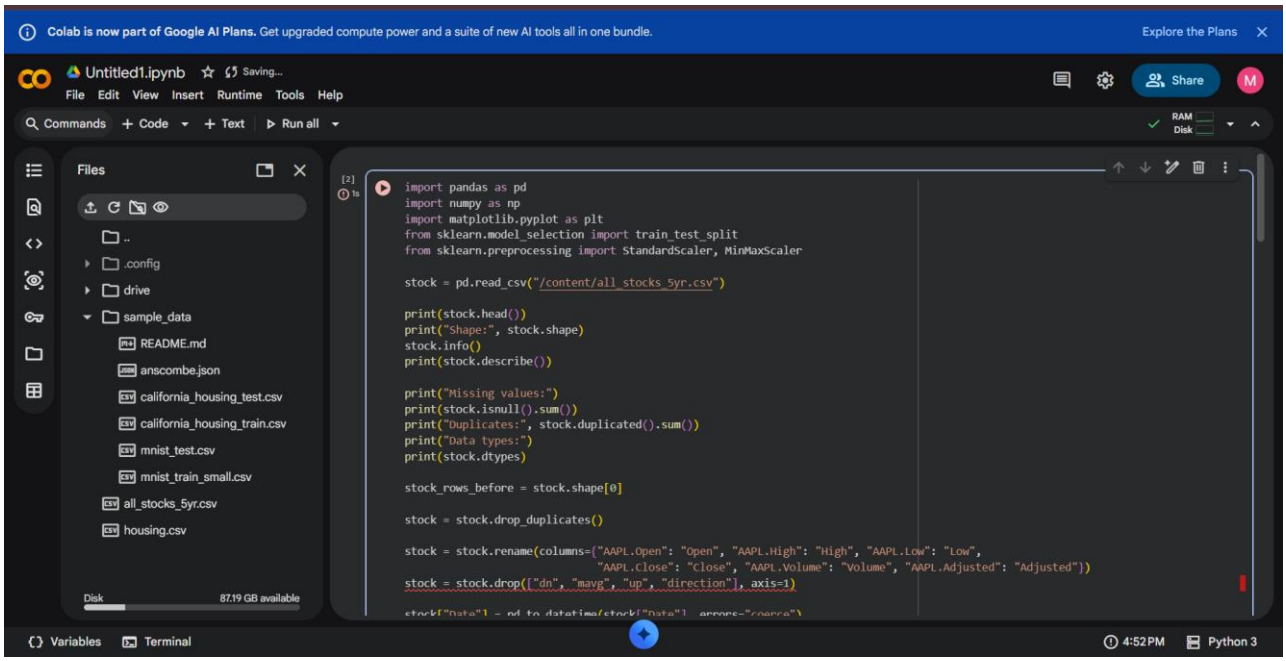

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RAM Disk
Skew before log: 0.8095011324315347
Skew after log: -0.3075432137708497
House rows before: 20640 after: 19311
House train: (15448, 15) test: (3863, 15)
longitude latitude housing_median_age total_rooms total_bedrooms \
3302 -1.510529 1.519411 -1.008321 -0.439389 -0.357941
17020 -1.360975 0.859230 0.513937 0.081182 -0.228516
1377 -1.256288 1.096337 0.674174 -0.400300 -0.477953
11720 -0.269235 1.663536 -0.607727 0.180513 0.075044
11971 1.056805 -0.767978 -0.207133 0.197529 0.131520
population households median_income rooms_per_household \
3302 -0.771436 -0.522975 -1.484212 0.077214
17020 -0.255902 -0.257588 1.319638 0.668654
1377 -0.280620 -0.409606 0.221065 -0.115513
11720 -0.936513 -0.912038 -0.319858 6.221487
11971 0.241976 0.121169 -0.574547 0.083933
bedrooms_per_room population_per_household ocean_proximity_INLAND \
3302 0.290375 -0.177943 True
17020 -0.932644 -0.027506 False
1377 -0.367215 0.036387 False
11720 -0.425115 -0.073640 True
11971 -0.335316 0.013225 True
ocean_proximity_ISLAND ocean_proximity_NEAR BAY \
3302 False False
17020 False False
1377 False True
11720 False False
11971 False False
Variables
Terminal
4:45 PM
Python 3

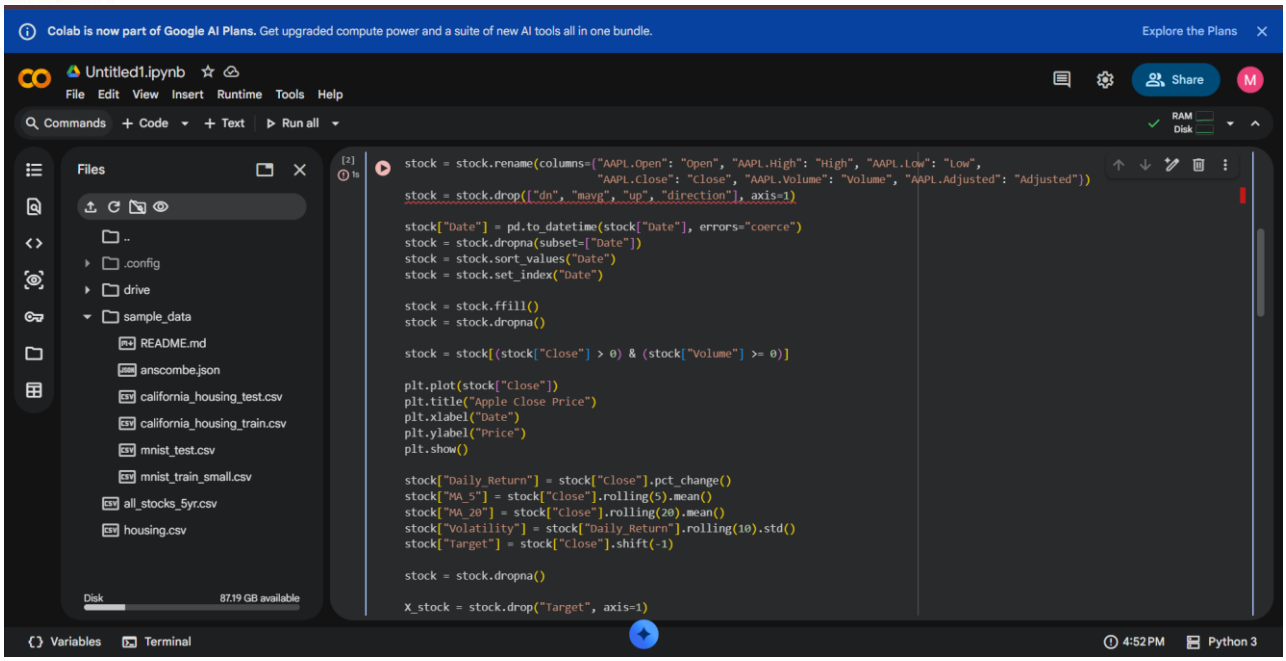

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RAM Disk
11720 -0.936513 -0.912038 -0.319858 6.221487
11971 0.241976 0.121169 -0.574547 0.083933
bedrooms_per_room population_per_household ocean_proximity_INLAND \
3302 0.290375 -0.177943 True
17020 -0.932644 -0.027506 False
1377 -0.367215 0.036387 False
11720 -0.425115 -0.073640 True
11971 -0.335316 0.013225 True
ocean_proximity_ISLAND ocean_proximity_NEAR BAY \
3302 False False
17020 False False
1377 False True
11720 False False
11971 False False
ocean_proximity_NEAR OCEAN
3302 False
17020 True
1377 False
11720 False
11971 False
Variables
Terminal
4:45 PM
Python

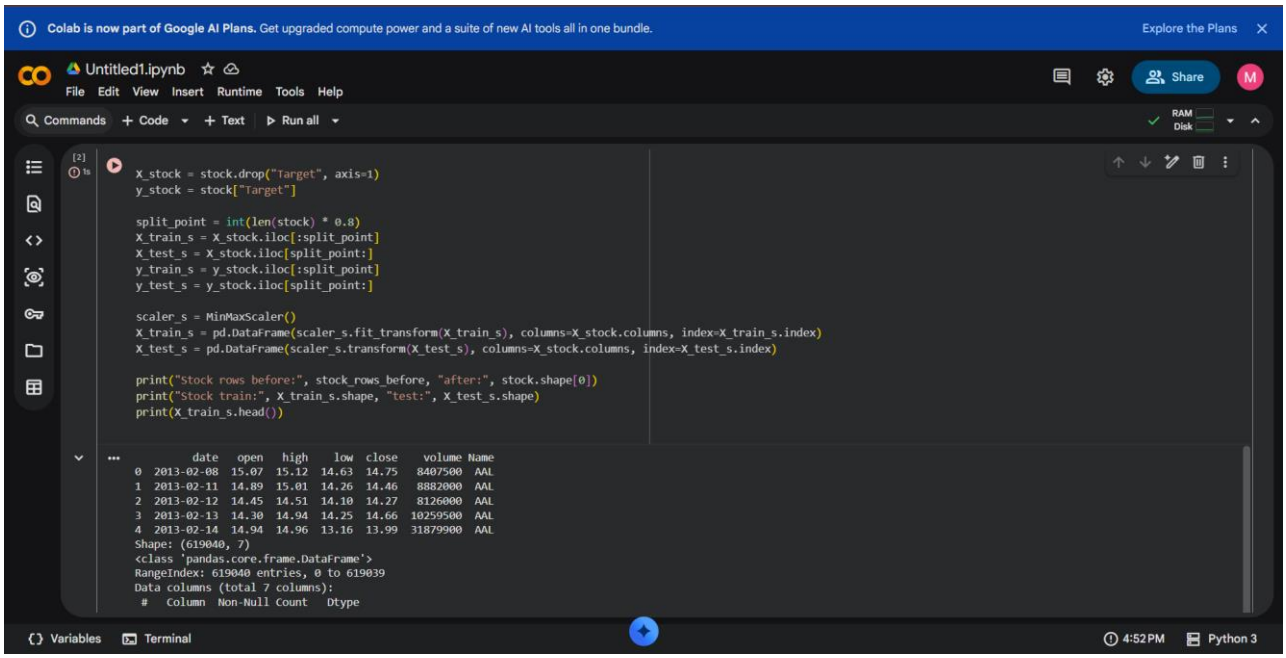

e4926fc8d072f2b4b4.gradio.live
New Chrome available
House Price Prediction
Linear Regression on California Housing dataset. R2 Score = 0.607
Longitude
-122.23
Latitude
37.88
Housing Median Age
41
Total Rooms
880
Total Bedrooms
129
Population
322
Households
126
Median Income (in 10k)
Result
Predicted House Price: $390,372.44
Flag

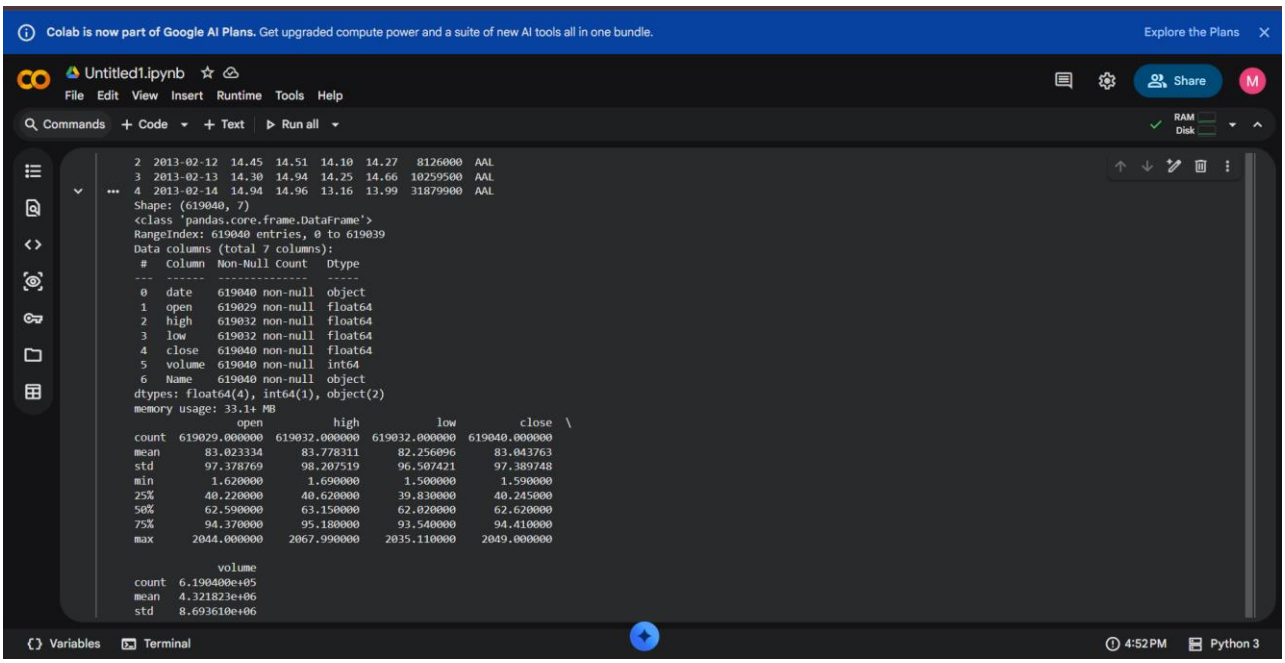


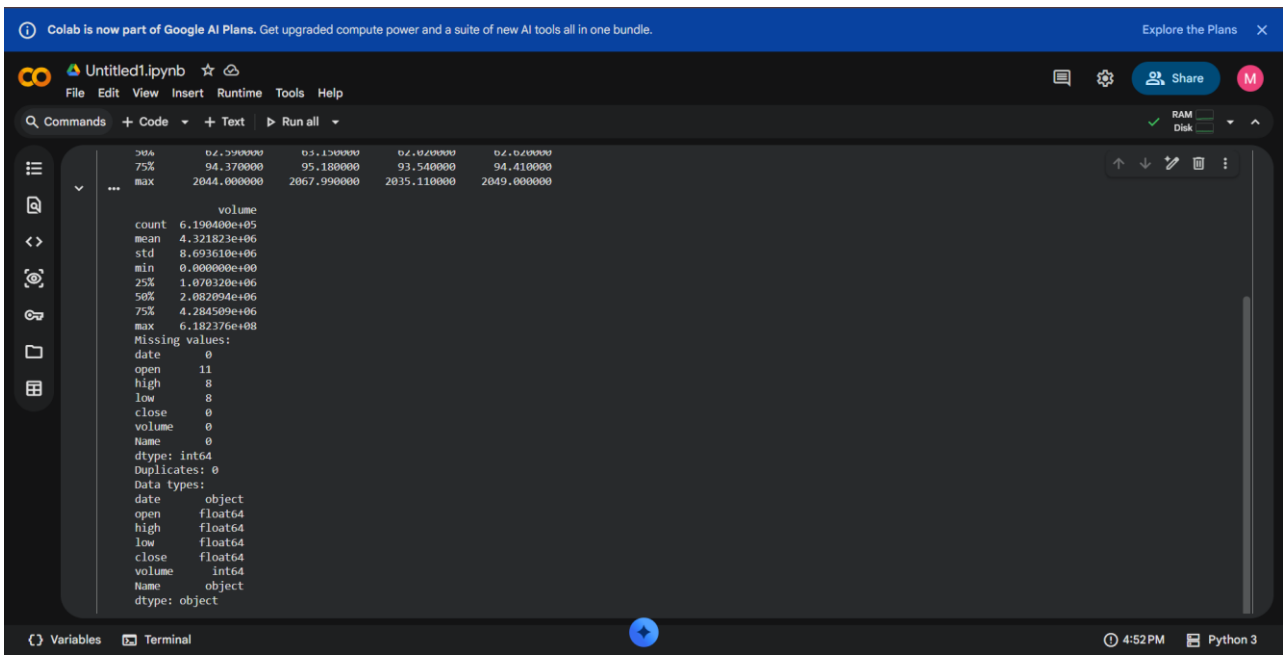
Stock Market Trend Analysis System

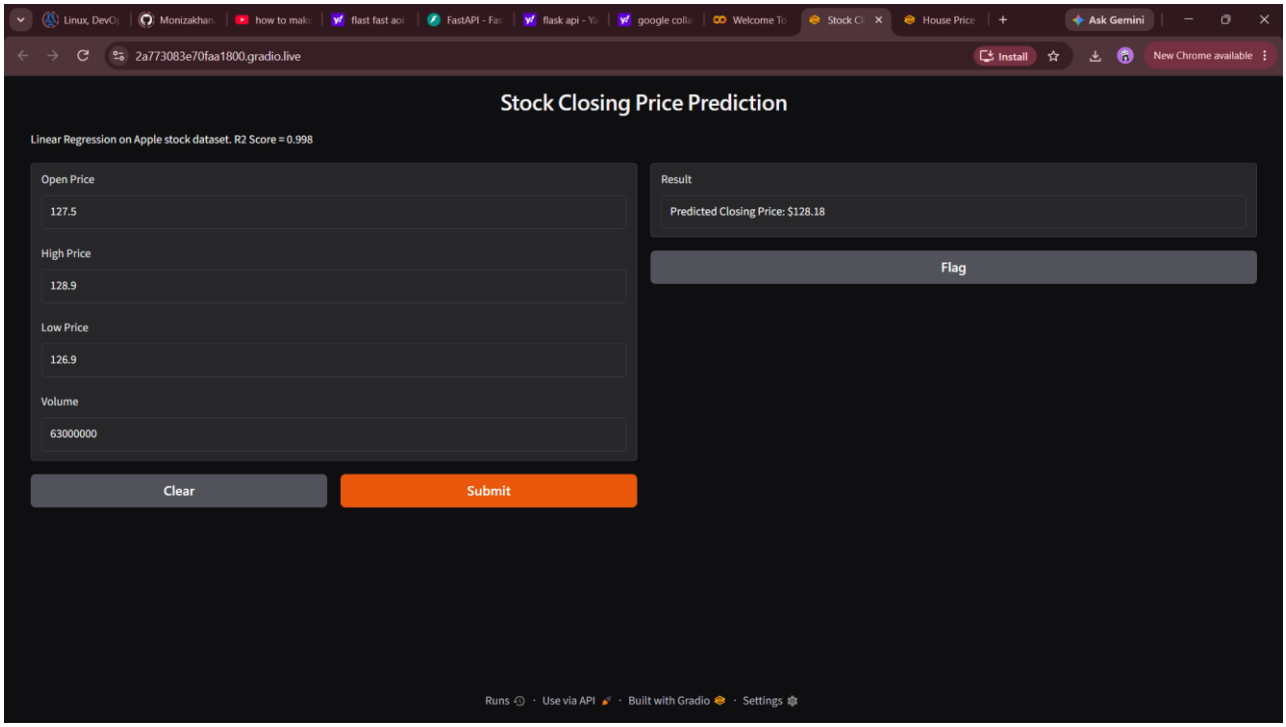



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Files
.config
drive
sample_data
README.md
anscombe.json
california_housing_test.csv
california_housing_train.csv
mnist_test.csv
mnist_train_small.csv
all_stocks_5yr.csv
housing.csv
Disk 87.19 GB available
stock = stock.rename(columns={"AAPL.Open": "Open", "AAPL.High": "High", "AAPL.Low": "Low",
"AAPL.Close": "Close", "AAPL.Volume": "Volume", "AAPL.Adjusted": "Adjusted"})
stock = stock.drop(["dn", "mavg", "up", "direction"], axis=1)
stock["Date"] = pd.to_datetime(stock["Date"], errors="coerce")
stock = stock.dropna(subset=["Date"])
stock = stock.sort_values("Date")
stock = stock.set_index("Date")
stock = stock.ffill()
stock = stock.dropna()
stock = stock[(stock["Close"] > 0) & (stock["Volume"] >= 0)]
plt.plot(stock["Close"])
plt.title("Apple Close Price")
plt.xlabel("Date")
plt.ylabel("Price")
plt.show()
stock["Daily_Return"] = stock["Close"].pct_change()
stock["MA_5"] = stock["Close"].rolling(5).mean()
stock["MA_20"] = stock["Close"].rolling(20).mean()
stock["Volatility"] = stock["Daily_Return"].rolling(10).std()
stock["Target"] = stock["Close"].shift(-1)
stock = stock.dropna()
X_stock = stock.drop("Target", axis=1)
Variables Terminal
4:52PM Python 3


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X_stock = stock.drop("Target", axis=1)
y_stock = stock["Target"]
split_point = int(len(stock) * 0.8)
X_train_s = X_stock.iloc[:split_point]
X_test_s = X_stock.iloc[split_point:]
y_train_s = y_stock.iloc[:split_point]
y_test_s = y_stock.iloc[split_point:]
scaler_s = MinMaxScaler()
X_train_s = pd.DataFrame(scaler_s.fit_transform(X_train_s), columns=X_stock.columns, index=X_train_s.index)
X_test_s = pd.DataFrame(scaler_s.transform(X_test_s), columns=X_stock.columns, index=X_test_s.index)
print("Stock rows before:", stock_rows_before, "after:", stock.shape[0])
print("Stock train:", X_train_s.shape, "test:", X_test_s.shape)
print(X_train_s.head())
date open high low close volume Name
0 2013-02-08 15.07 15.12 14.63 14.75 8407500 AAL
1 2013-02-11 14.89 15.01 14.26 14.46 8882000 AAL
2 2013-02-12 14.45 14.51 14.10 14.27 8126000 AAL
3 2013-02-13 14.30 14.94 14.25 14.66 10259500 AAL
4 2013-02-14 14.94 14.96 13.16 13.99 31879900 AAL
Shape: (619040, 7)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 619040 entries, 0 to 619039
Data columns (total 7 columns):
Column Non-Null Count Dtype
Variables Terminal
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RAM Disk
2 2013-02-12 14.45 14.51 14.10 14.27 8126000 AAL
3 2013-02-13 14.30 14.94 14.25 14.66 10259500 AAL
4 2013-02-14 14.94 14.96 13.16 13.99 31879900 AAL
Shape: (619040, 7)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 619040 entries, 0 to 619039
Data columns (total 7 columns):
Column Non-Null Count Dtype
0 date 619040 non-null object
1 open 619029 non-null float64
2 high 619032 non-null float64
3 low 619032 non-null float64
4 close 619040 non-null float64
5 volume 619040 non-null int64
6 Name 619040 non-null object
dtypes: float64(4), int64(1), object(2)
memory usage: 33.1+ MB
open high low close \
count 619029.000000 619032.000000 619032.000000 619040.000000
mean 83.023334 83.778311 82.256096 83.043763
std 97.378769 98.207519 96.507421 97.389748
min 1.620000 1.690000 1.500000 1.590000
25% 40.220000 40.620000 39.830000 40.245000
50% 62.590000 63.150000 62.020000 62.620000
75% 94.370000 95.180000 93.540000 94.410000
max 2044.000000 2067.990000 2035.110000 2049.000000
volume
count 6.190400e+05
mean 4.321823e+06
std 8.693610e+06
Variables Terminal
4:52 PM Python 3


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RAM Disk
75% 94.370000 95.180000 93.540000 94.410000
max 2044.000000 2067.990000 2035.110000 2049.000000
volume
count 6.190400e+05
mean 4.321823e+06
std 8.693610e+06
min 0.000000e+00
25% 1.070320e+06
50% 2.082094e+06
75% 4.284509e+06
max 6.182376e+08
Missing values:
date 0
open 11
high 8
low 8
close 0
volume 0
Name 0
dtype: int64
Duplicates: 0
Data types:
date object
open float64
high float64
low float64
close float64
volume int64
Name object
dtype: object
Variables Terminal
4:52 PM Python 3


2a773083e70faa1800.gradio.live
Install
New Chrome available
Stock Closing Price Prediction
Linear Regression on Apple stock dataset. R2 Score = 0.998
Open Price
127.5
High Price
128.9
Low Price
126.9
Volume
63000000
Clear
Submit
Result
Predicted Closing Price: $128.18
Flag
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